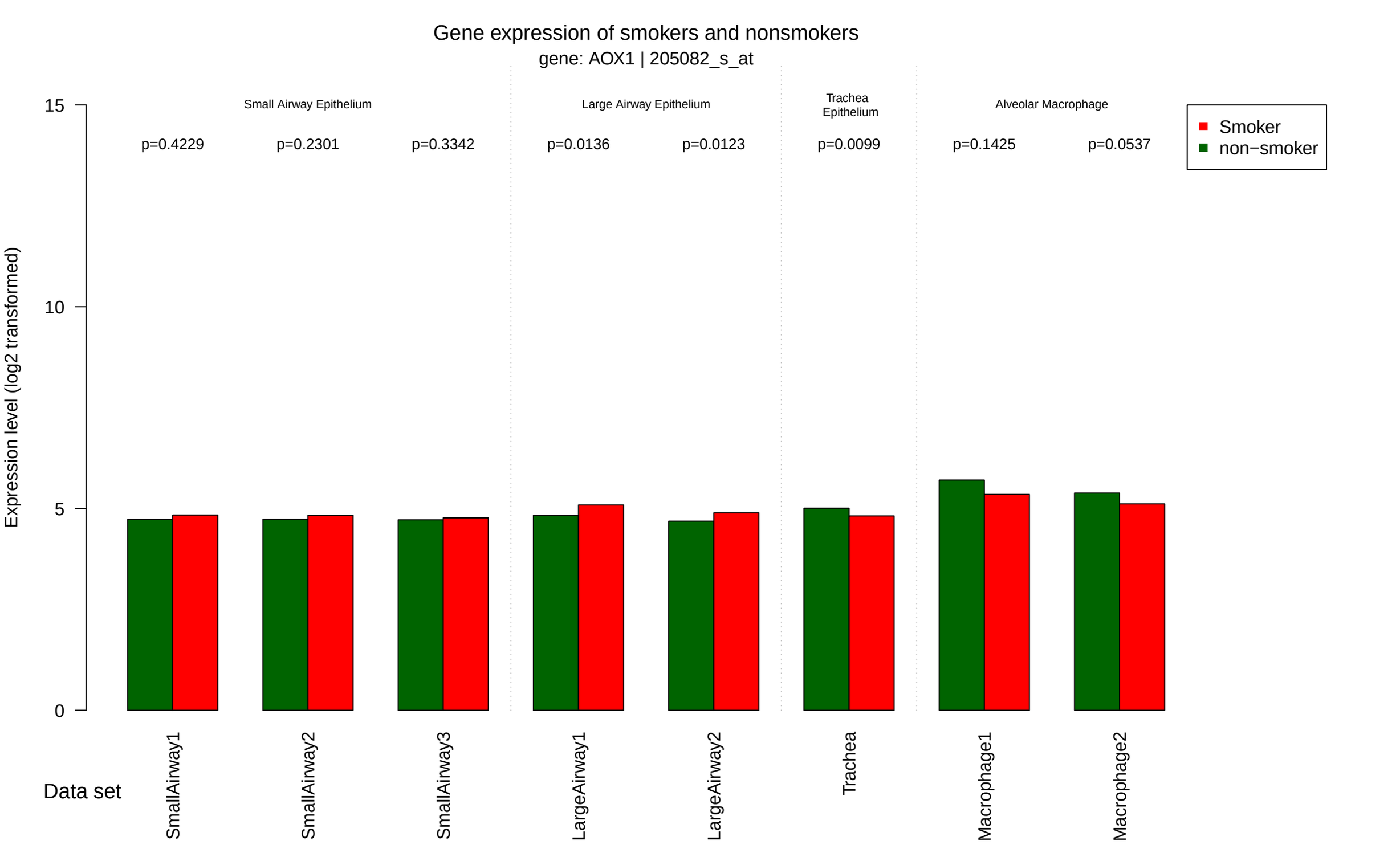
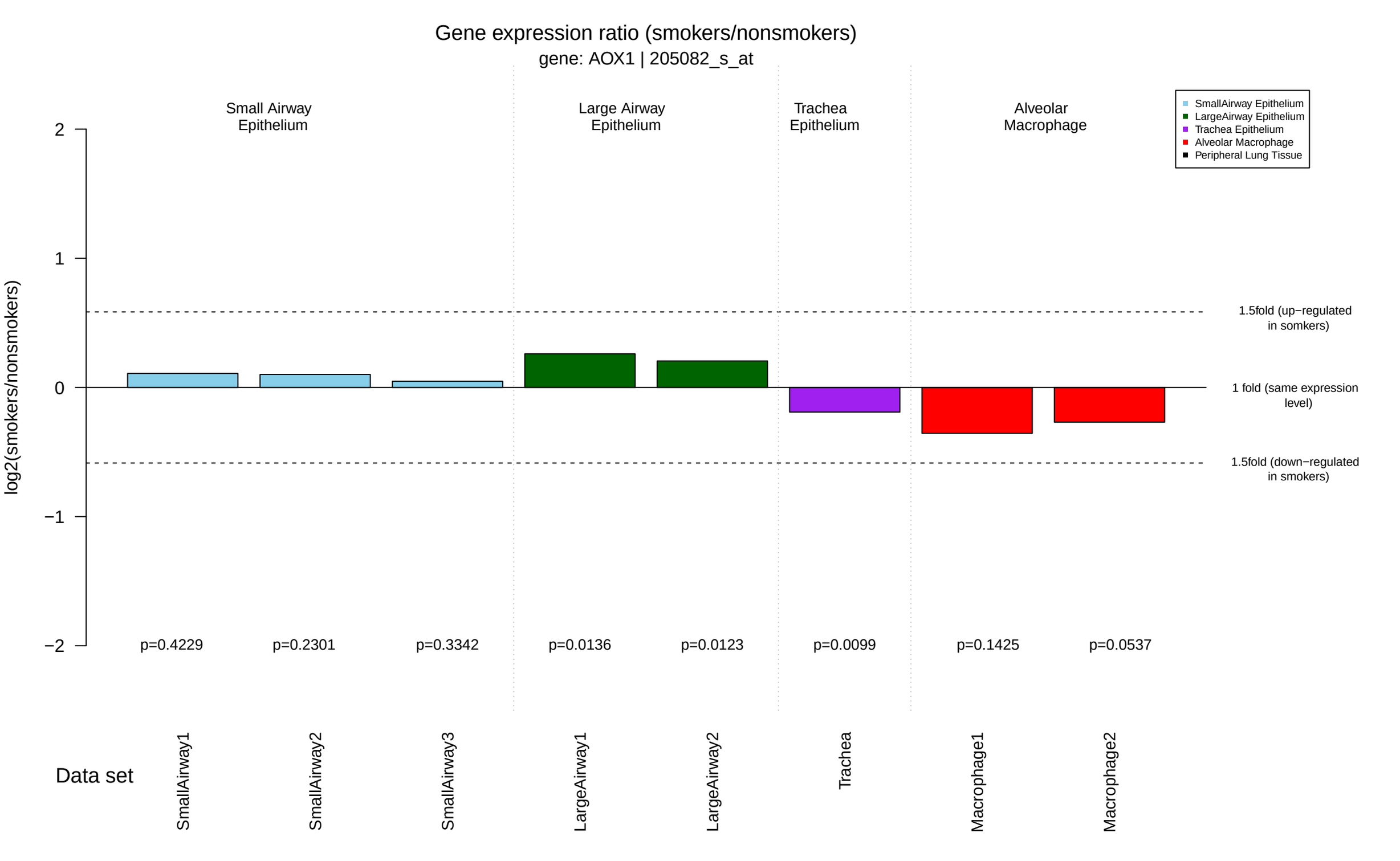
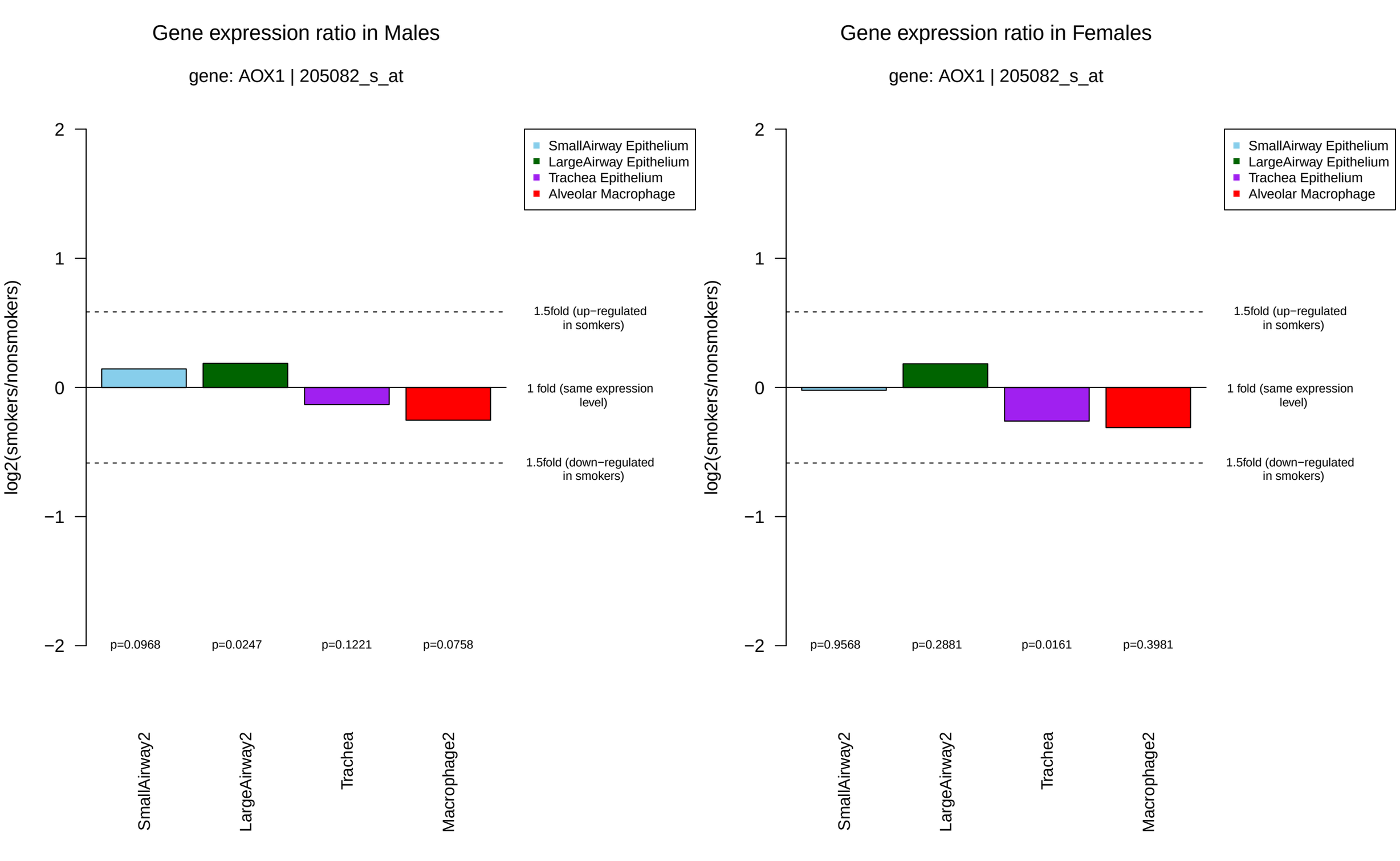
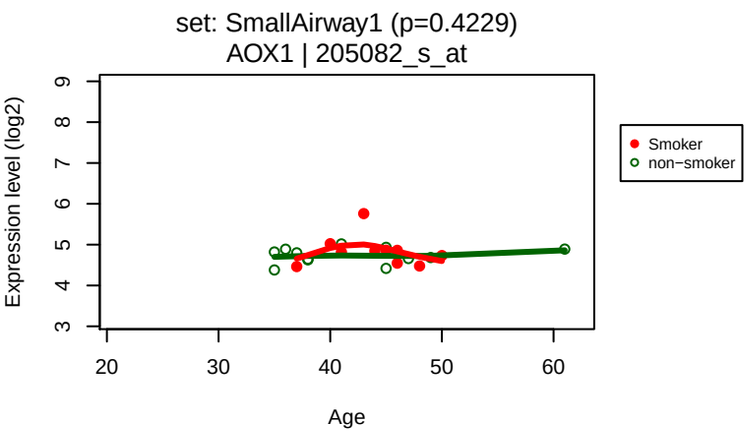
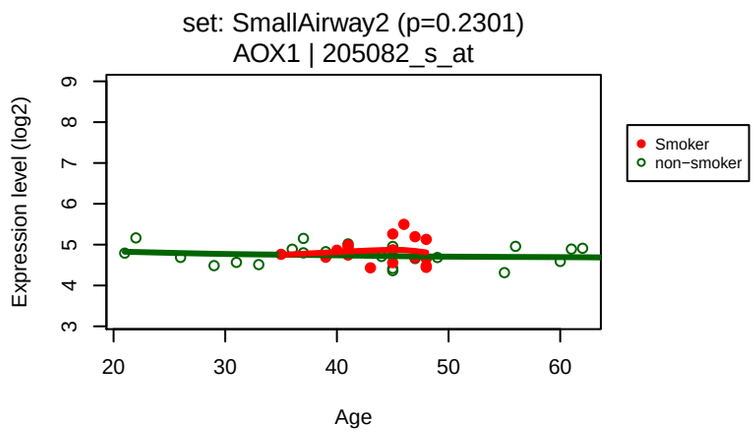
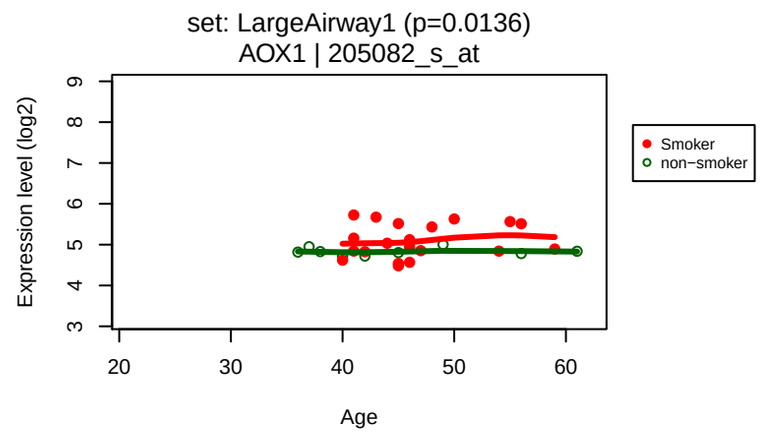

Gene expression of smokers and nonsmokers
gene: AOX1 | 205082_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4229
p=0.2301
p=0.3342
p=0.0136
p=0.0123
p=0.0099
p=0.1425
p=0.0537
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

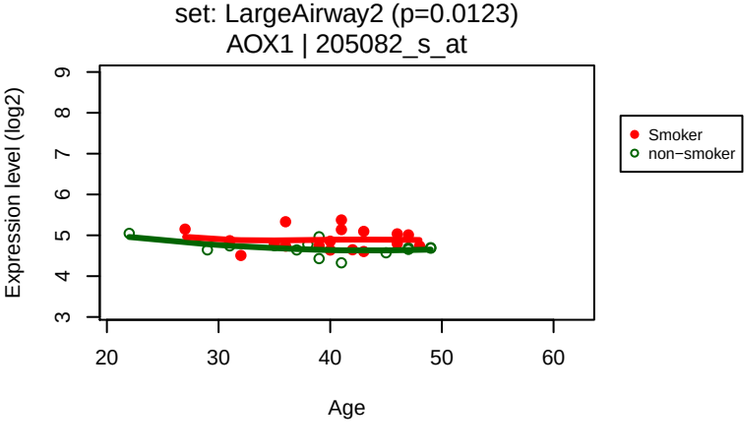

Gene expression ratio (smokers/nonsmokers)
gene: AOX1 | 205082_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4229
p=0.2301
p=0.3342
p=0.0136
p=0.0123
p=0.0099
p=0.1425
p=0.0537
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

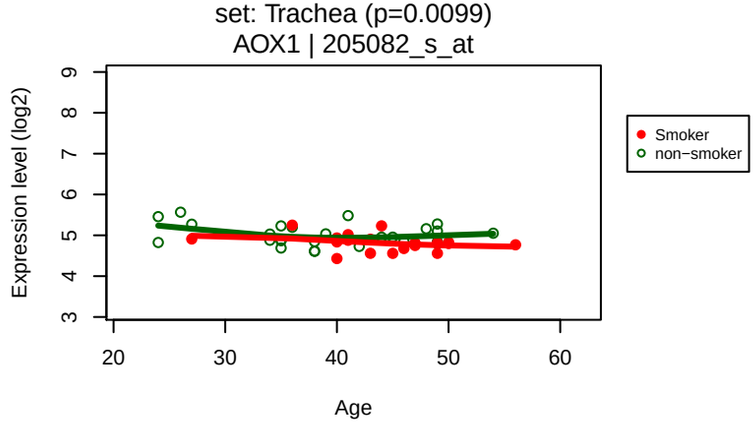

Gene expression ratio in Males
gene: AOX1 | 205082_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0968
p=0.0247
p=0.1221
p=0.0758
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AOX1 | 205082_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9568
p=0.2881
p=0.0161
p=0.3981
SmallAirway2
LargeAirway2
Trachea
Macrophage2

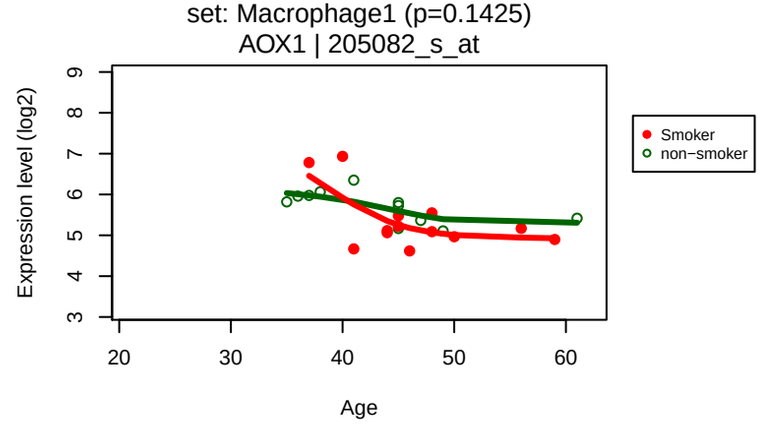

set: SmallAirway1 (p=0.4229)
AOX1 | 205082_s_at
Expression level (log2)
Age
Smoker
non−smoker

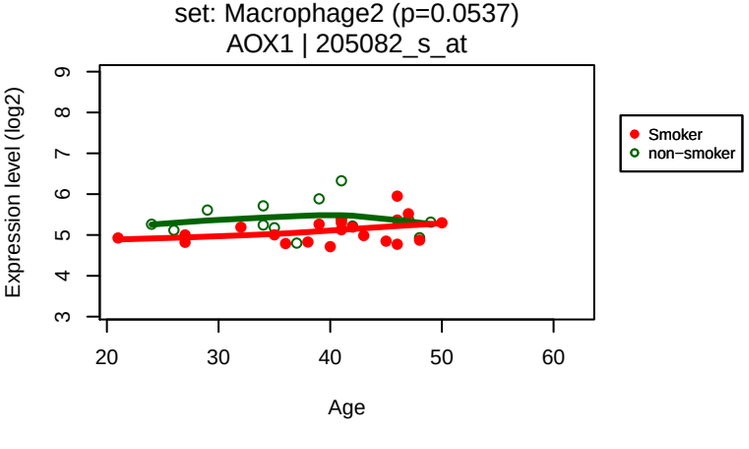

set: SmallAirway2 (p=0.2301)
AOX1 | 205082_s_at
Expression level (log2)
Age
Smoker
non−smoker

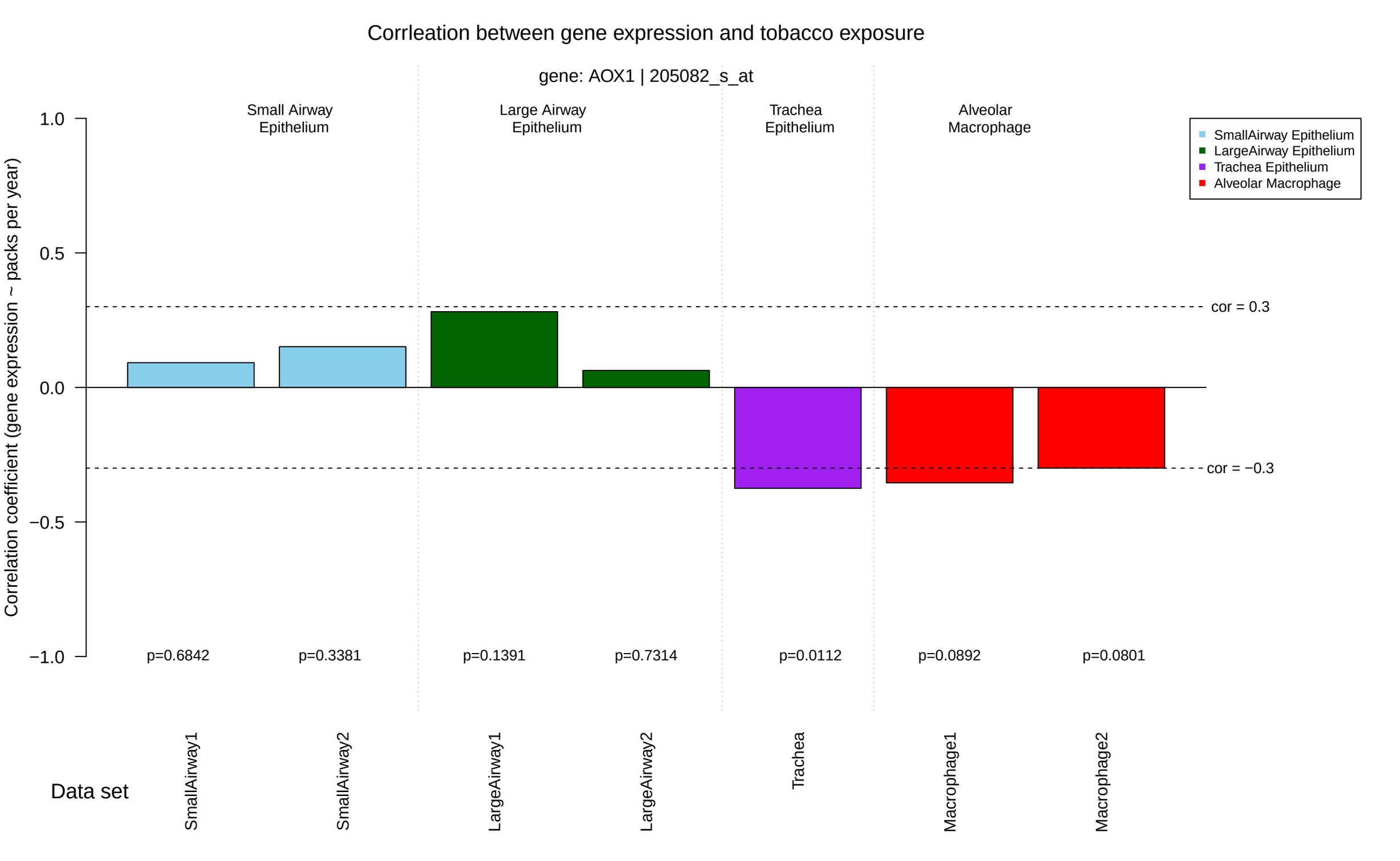

set: LargeAirway1 (p=0.0136)
AOX1 | 205082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0123)
AOX1 | 205082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0099)
AOX1 | 205082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1425)
AOX1 | 205082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0537)
AOX1 | 205082_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AOX1 | 205082_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6842
p=0.3381
p=0.1391
p=0.7314
p=0.0112
p=0.0892
p=0.0801
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2